LERAY SPECTRAL SEQUENCE FOR THE DERIVED PUSHFORWARD OF THE QC CATEGORY ON ALGEBRAIC STACKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Leray spectral sequence for ringed sites. Stacks Project, Cohomology on Sites, Leray spectral sequence (as specialized in sites-cohomology). Provides the two spectral sequences being compared.
- Kernel and cokernel of the adjunction map are parasitic. Stacks Project, Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-adjoint-kernel-parasitic. Identifies the comparison map's kernel and cokernel.
- Cohomology of parasitic modules vanishes. Stacks Project, Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-pushforward-parasitic. Ensures the comparison map induces isomorphisms on cohomology.

Source reference. Original reference 0732: sources/stacks-chow/tags.txt, source lines 9046–9046. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0779: sources/stacks-chow/tags.txt, source lines 9194–9194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0775: sources/stacks-chow/tags.txt, source lines 9190–9190. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a quasi-compact and quasi-separated morphism of algebraic stacks. Let $\mathcal{F}$ be a quasi-coherent sheaf on $\mathcal{X}$. Then there exists a spectral sequence with E_2-page*

$$E_2^{p,q} = H^p(\mathcal{Y}, R^q f_{QCoh,*}\mathcal{F})$$

converging to $H^{p+q}(\mathcal{X}, \mathcal{F})$.

Proof. By Cohomology on Sites, Lemma 0732 the Leray spectral sequence with

$$E_2^{p,q} = H^p(\mathcal{Y}, R^q f_*\mathcal{F})$$

converges to $H^{p+q}(\mathcal{X}, \mathcal{F})$. The kernel and cokernel of the adjunction map

$$R^q f_{QCoh,*}\mathcal{F} \longrightarrow R^q f_*\mathcal{F}$$

are parasitic modules on $\mathcal{Y}$ (Lemma 0779) hence have vanishing cohomology (Lemma 0775). It follows formally that $H^p(\mathcal{Y}, R^q f_{QCoh,*}\mathcal{F}) = H^p(\mathcal{Y}, R^q f_*\mathcal{F})$ and we win. □

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